



JOATHAM JACOB

Energy and Decarbonisation Engineer

Industrial energy efficiency · Marginal abatement cost modelling · Energy management systems (ISO 50001)

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Italian residence permit, work-authorised in Italy · Open to relocation

PROFESSIONAL EXPERIENCE

ESG Supply Chain & International Project Sourcing Intern – Industrious Global Technologies, Milan Feb 2025 – Aug 2025

I built the company's 2024 Scope 1 and 2 emissions inventory from primary consumption data under the GHG Protocol, with formal quality controls on internal and supplier data. I also screened international suppliers against ESG criteria and built the KPI reporting used by management.

ESG Measurement Intern — Omtra S.r.l Società Benefit, Milan, Italy Nov 2023 – May 2024

I carried out a cradle-to-grave product LCA for Cartier and calculated Scope 1–3 emissions for client companies from data collected on site. For Busatti, a Tuscan textile maker, I built its carbon footprint across 2 production sites. I also contributed to its first statutory Impact Report and I supported its B Impact Assessment to 89.9 points (threshold 80; industry average 75.5), and supported B Corp and EcoVadis verification across clients.

ESG Analyst (Remote) — Earth Academy Global, Amstelveen, Netherlands May 2021 – Nov 2022

I analysed corporate sustainability disclosures against five international frameworks (GRI, SASB, TCFD, Integrated Reporting and India's BRSR) for clients across Europe, the Middle East and India. I also worked on the firm's ESG training programmes for corporate clients. With ISOS Group, I supported GRI training in Dubai and an ESG masterclass for Al-Dabbagh Group in Saudi Arabia, and I supported training for sustainability professionals at Tanmiah Food Company. I joined as an intern focused on the then-new BRSR framework and moved into the full-time analyst role.

TECHNICAL PROJECTS AND FIELD WORK

Industrial decarbonisation assessment, Acciaieria Arvedi, Cremona (2026)

Rebuilt the Scope 1 baseline at 434,844 tCO₂ (EU ETS, CY2023) and showed that its net-zero claim rests on 3 accounting instruments and 1 physical reduction.

- Costed a 2030 MACC: 192,494 tCO₂/yr (44.3%) at a modelled €58.0M capex, €11.44 per tonne of steel. The capex and opex are estimates, not quotes.
- Found that 59% of the abatement depends on ~36 kt/yr of biochar, and ETS and CBAM costs rise from €6.9M to €32.6M/yr.

Energy monitoring pipeline (Python, Power BI, 2026): degree-day regression baseline with deviation flagging for ISO 50001-style reporting.

Web tools (jacob-hub-esg.github.io, 2026): 4 live tools, covering a MACC and finance calculator, the BuildingCarbon benchmark, CSRD readiness (10 ESRS) and a CO₂ calculator.

Field audits (B.Tech): energy, water and safety audits of 8 railway stations in Bihar over 25 days, plus a biogas plant performance evaluation in Palakkad.

EDUCATION

M.Sc. Environmental Change and Global Sustainability — Università degli Studi di Milano, Italy (2022–2025)

- Classe LM-75. Full academic scholarship. Final grade 100/110. Taught entirely in English.
- **Thesis:** Emission Reduction Commitments in Global Agriculture: A Comparative Analysis of Corporate Climate Disclosures.

B.Tech Energy and Environmental Engineering — Tamil Nadu Agricultural University, India (2016–2021)

- **Thesis:** Design and Fabrication of a Solar Powered Trash Compactor (published, TNAU 2020).

SKILLS

- **Energy & decarbonization:** energy auditing · ISO 50001 (baselines, EnPIs, monitoring) · MACC · industrial energy systems (motors and VFDs, compressed air, steam, process heat) · heat electrification · waste-heat recovery · M&V
- **Analysis & finance:** NPV · IRR · LCOE · levelised cost of abatement · payback · scenario and sensitivity analysis · regression baselining
- **Carbon & regulation:** GHG Protocol Scope 1–3 · ISO 14064-1/-2 · LCA · EU ETS · CBAM · CSRD/ESRS · EU Taxonomy · EPBD · SBTi · B Corp · EcoVadis · GRI · SASB · TCFD
- **Tools:** Excel (advanced: Power Query, pivots, modelling) · Power BI · Python (pandas) · React · Tableau · PV Syst · AutoCAD · QGIS · Articulate Storyline
- **Languages:** English (C1, IELTS 7.5) · Italian (B1) · Malayalam (native) · Tamil · Hindi

CERTIFICATIONS

- GHG 202: Basics of Project-Level GHG Accounting, GHG Management Institute (in progress, 2026)
- GHG Accounting: Corporate Standard, GHG Protocol (2026)
- ISO 50001 Energy Management Systems, Coursera (2026); willing to pursue ISO 50001 Lead Auditor certification
- IGBC Accredited Professional
- Earth Academy Practitioner: GRI, TCFD, Integrated Reporting, SDGs